

Future Trends Shaping Reality

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צתידים בחינוך

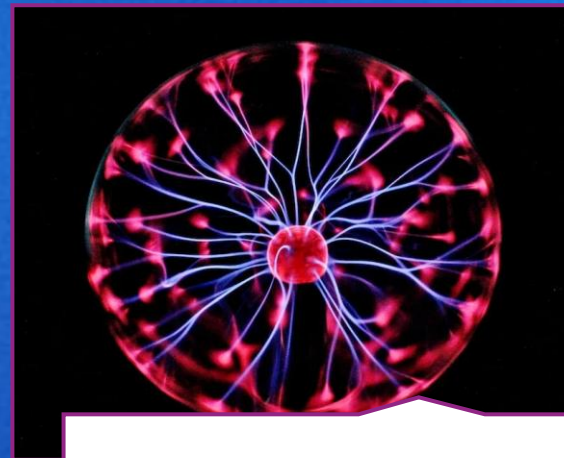
The World We Live In – VUCA



Volatility



Uncertainty



Complexity



Ambiguity

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The Era We Live In



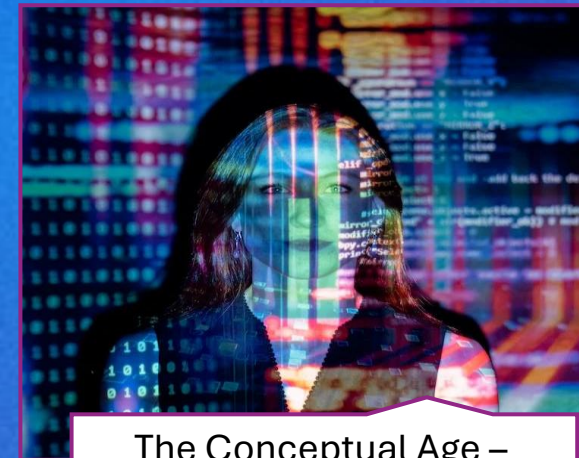
The Agricultural Age – Land



The Industrial Age – Capital



The Information Age –
Knowledge



The Conceptual Age –
Intuition, Creativity,
Imagination

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Who Are the Survivors? (Darwin)

National Geographic



“It is not the strongest
or the most intelligent



but the one most
responsive to change.”

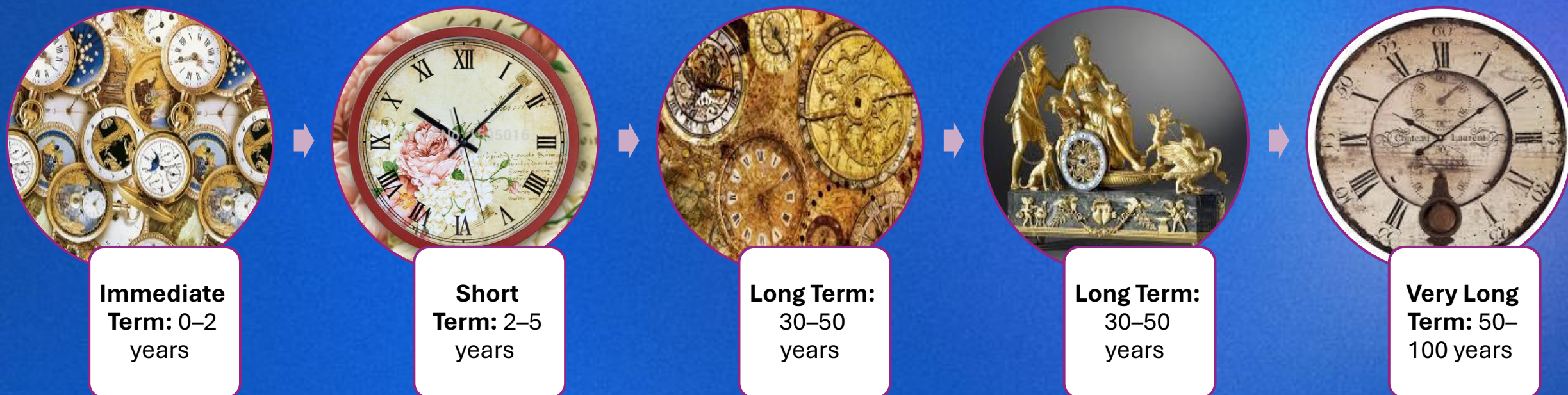
Can We Predict the Future?

- *Futures research is an oxymoron... since one cannot research what has not yet occurred!
- *Futurism is also not prophecy... its goal is to understand changing reality through:
 - *Identifying trends from the **past**
 - *And promoting actions in the **present**
 - *In order to prepare for the **future**



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Future Time Horizons

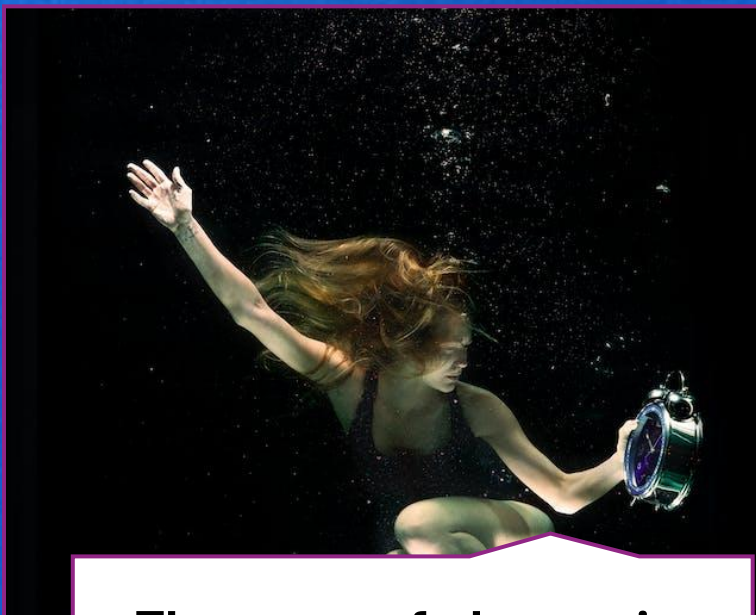


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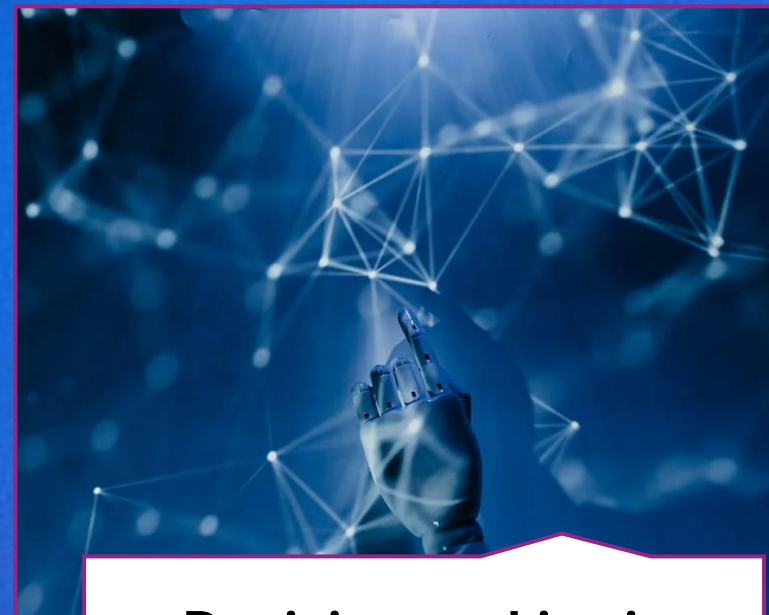
Challenges of Future Thinking



**Human thinking is linear
and biased**



**The pace of change is
accelerating**



**Decision-making is
becoming more complex**

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Futures: From Probable to Wild Cards



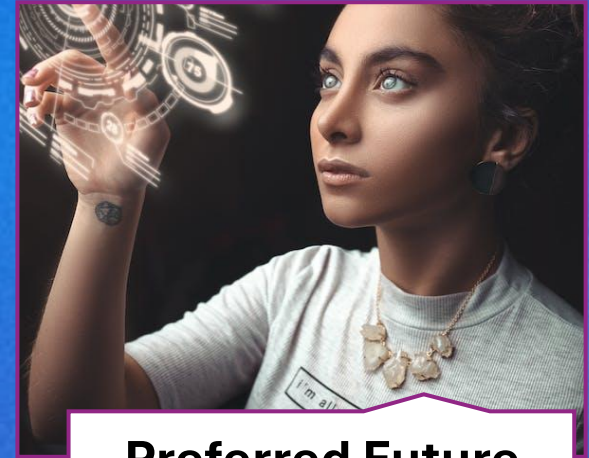
**Probable (Classic)
Future – (~60%)**



**Possible Future –
(~40–50%)**



**Wild Card Future –
(~30%)**



**Preferred Future
(Future We Shape)**

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The STEEP Dimensions



Social



Technology



Economy



Environment



Politics

"Meta-Trends" Changing the Nature of Humanity

1. Convergence as a Driver of Innovation

- Integration of AI, biotechnology, quantum and nanotechnology
- Creation of new capabilities
- Acceleration of human progress

2. Human–Machine Interfaces

- Blurring boundaries between humans and technology
- Brain–computer interfaces, neural implants and smart devices
- Expansion of sensory and cognitive abilities

3. Human Enhancement & Bio-Cognitive Gaps

- Enhancement of human abilities through technology
- Gene editing and cognitive/physical augmentation
- Possible gaps between enhanced and non-enhanced humans

4. Life in a Post-Work World

- Automation and AI reduce the need for traditional work
- More space for meaning, learning and creativity
- New questions about human identity and purpose

Convergence



The integration of diverse domains into a unified, adaptive system, bridging education, technology, and the social sciences.

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Every significant innovation of the last 150 years stems from technological convergence



19th Century
Telephone (1876)
Radio (1895)



20th Century
Television (1927)
Talking films (1927)
Computer (1940)
Internet (1969)
Mobile phone (1973)
GPS (1978)
Digital media player



21st Century
Social networks (2000)
Smartphone (2007)
Blockchain (2008)
Autonomous vehicles (2010)



Since 2010
Smartwatch
3D printing
Drones
Touch tablets
Nanotechnologies
Mixed reality
Solid-state batteries

Artificial Intelligence as an Enabler... but this is only the beginning...

Enhanced Experience AI

Expands learning through multimodal and hybrid environments.

- **Multimodal AI** – integrates text, image, audio and video
- **Phyigital AI** – connects physical and digital realities via AR/VR

Action & Creation AI

Acts, creates & collaborates with humans

- **Agentic AI** – plans and acts autonomously
- **Physical AI** – operates in the physical world
- **Creative AI** – generates original content
- **Collaborative AI** – enables human-machine teamwork

Human-Centric AI

Interprets human states: emotion, cognition, senses and body.

- **Emotional AI** – recognizes emotions
- **Cognitive AI** – supports thinking and problem-solving
- **Sensory AI** – reads physical and emotional signals

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From a Nomadic Society (1.0) to Society 5.0



Society 1.0



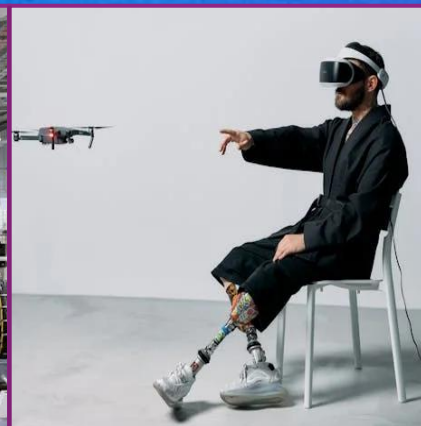
Society 2.0



Industry 3.0



**& Industry 4.0
Society 4.0**



Society 5.0

The Human-Machine Interface

*Blurring the Boundaries Humans are not merely users of technology – they integrate into it and become part of it



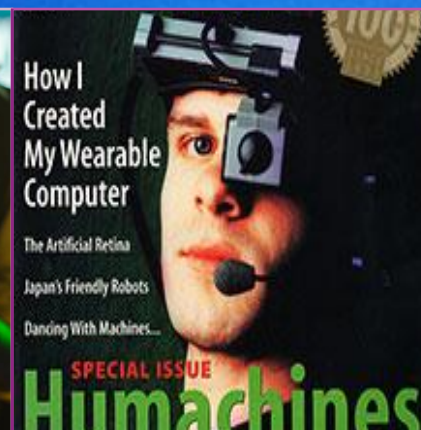
**Brain–
Computer
Interfaces (BCI)**



**Neural and
Subdermal
Implants**



Posthumanism



**Humachine –
Human–
Machine
Symbiosis**

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Human Enhancement and the Bio-Cognitive Gap



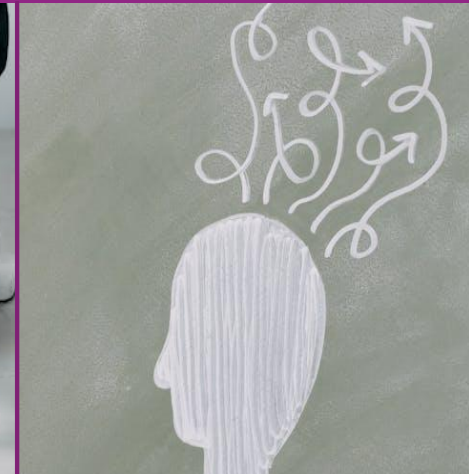
**Advanced
Gene Editing**



**Designer
Babies**



**Human
Enhancement**



**Risk of a Bio-
Cognitive
Divide**



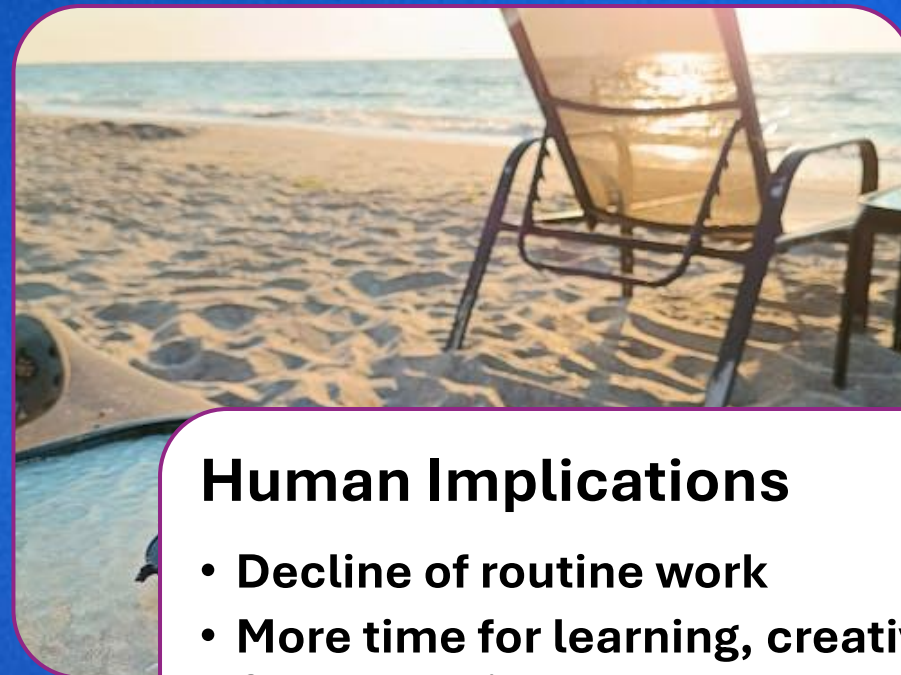
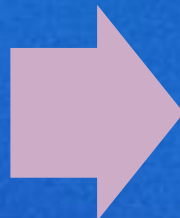
**“Enhanced”
vs. “Non-
Enhanced”**

Extending Life Expectancy in a Post-Work World



Technological Drivers

- **Abundant energy** – fusion & renewables
- **Biomanufacturing** – cellular agriculture, bioprinting
- **Automation, AI & robotics** – intelligent systems
- **Radical life extension** – treating aging as a disease



Human Implications

- **Decline of routine work**
- **More time for learning, creativity & community**
- **Redefining identity and value beyond work**
- **Equity challenges** – gaps between “enhanced” and others

The Dilemma

"Is the lecturer or teacher alone who should lead the educational process within their field of knowledge, or should an educational triangle be formed among the lecturer/teacher, the student/pupil and artificial intelligence?"

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There Is No Dilemma... There Is a New Educational Triangle

משולש ה-STEM החינוכי



- * AI does not replace the teacher, nor does it leave the learner alone, It creates a new learning system in which:
- * The teacher leads pedagogically
- * The learner acts and explores
- * Artificial intelligence expands, adapts, and accelerates the learning process

Implications for Education in General and STEM in Particular

*Transition to Lifelong Learning

*Academia

- *From "profession factory" to a space for thinking, identity, and adaptability
- *Anchor for scientific depth, interdisciplinary links, and innovation

*From Knowledge to Capabilities

- *Deep skills, creativity, critical thinking, complex problem-solving
- *STEM beyond disciplines: ethics, philosophy, complex questions

*Learning in a Changing World

- *Personalized, dynamic, multi-track learning
- *Self-directed learning and continuous renewal
- *Preparing learners for uncertainty, change, and working with AI

Challenges

*Thinking & Learning

- *Over-reliance on AI
- *Weakened independent thinking
- *Quick answers over deep learning

*Literacy & Gaps

- *Difficulty judging reliability, bias, and errors
- *AI and digital skill gaps between learners and lecturers

*Ethics & Responsibility

- *Blurred boundaries: who thinks, creates, and is accountable?
- *Privacy, transparency, and fairness in assessment

Opportunities

*Skills & Literacy

- *Core skills: critical thinking, creativity, complex problem-solving
- *Strong AI literacy: informed use, verification, understanding limits

*Learning & STEM

- *Personalized, self-paced learning
- *Expanded STEM via simulations, data, immersive environments

*Educators & Learners

- *More time for guidance and higher-order thinking
- *Independent, lifelong learners able to adapt and evolve

STEM – Beyond Technical Knowledge

- * Establishes cross-boundary thinking that connects scientific-technological knowledge, humanistic approaches, and diverse skills
- * Enables understanding of complex problems through the interplay between humans and technology
- * Contributes to understanding not just how technology works, but also what its implications are
- * Leads from training users to developing Socio-Technological Leaders, with moral responsibility, value judgment, and an understanding of human-machine relations



Thank You

